CRITIQUE OF THE WIDE VIEW & LINE FILTER

Computational Verification & Benchmark

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OUTLINE

1. Background
2. Math Verification
3. Benchmark Results
4. Angle Detection
5. Maritime Replication
6. Assessment

BACKGROUND

PAPERS UNDER REVIEW

Three publications (USC Mech. Eng.):

1. **M.S. Thesis** (2023) — WVF & LF
2. **IEEE OCEANS 2024** — Aquatic gradient computation
3. **IEEE OCEANS 2025** — Multi-scale edge determination

ONR Contract N00014-22-C-1011

Claims:

- Outperform Sobel, Prewitt, Canny
- Outperform DexiNed, TEED, EDTER
- $<1°$ angular accuracy
- Superior in low-SNR aquatic scenes
- ML is “unreliable and dangerous”

HOW THE WVF WORKS

2D Taylor expansion at each pixel:

$$f(x_i, y_i) \approx f^0 + f^0_x x + f^0_y y + \frac{f^0_{xx} x^2}{2} + \cdots$$

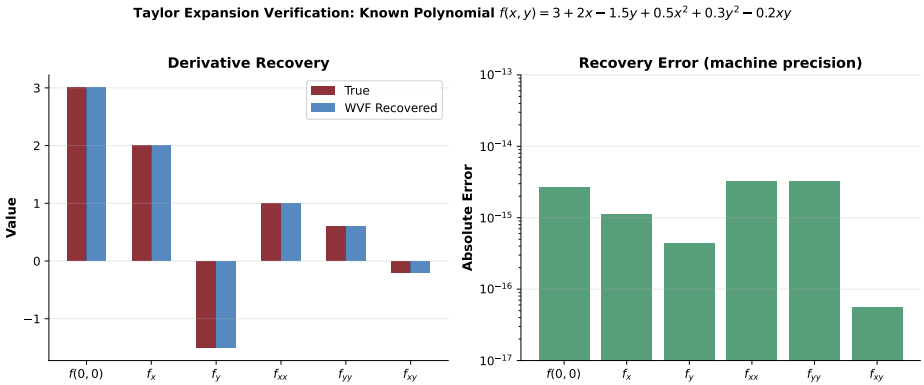
- N_p neighbors in circular region
- Least-squares solve per orientation
- Line Filter chains WVF along a line
- Papers use $N_p = 250$, $m = 14$

	Sobel	**WVF**
Pixels used	9	250
Kernel size	3×3	∼18px radius
Orientations	2 (x, y)	N_s (any)
Runtime	μs	seconds

28× more data per pixel—but only compared against 3×3.

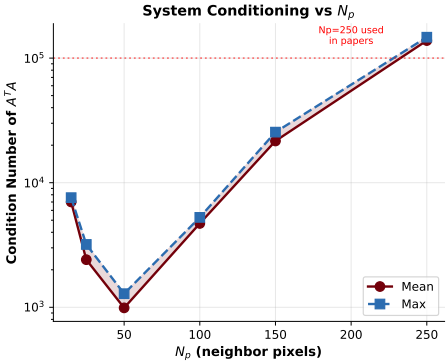
MATH VERIFICATION

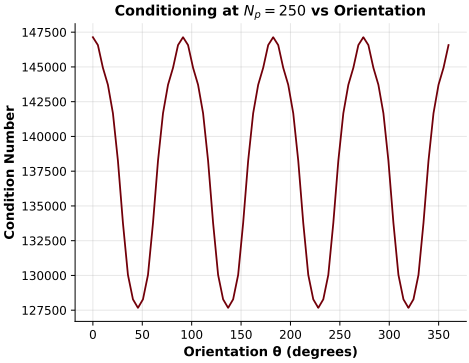
TAYLOR EXPANSION VERIFIED



All derivatives recovered to $\sim 10^{-15}$. **The math is sound.**

CONDITION NUMBER GROWTH

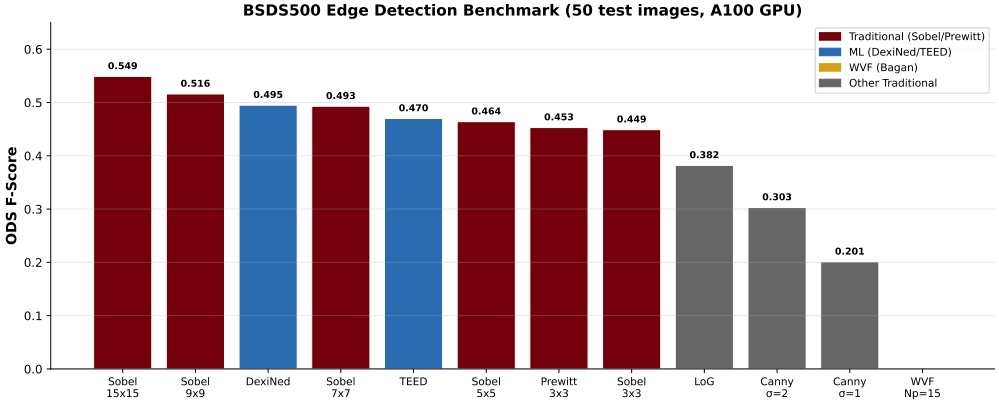




$\kappa(\mathbf{A}^T\mathbf{A}) \sim 10^5$ at $N_p = 250$. Not discussed in the publications.

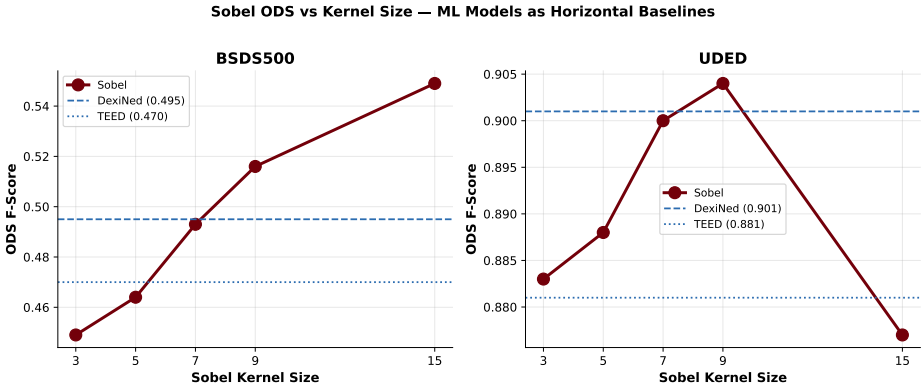
BENCHMARK RESULTS

BSDS500: LARGER SOBEL WINS



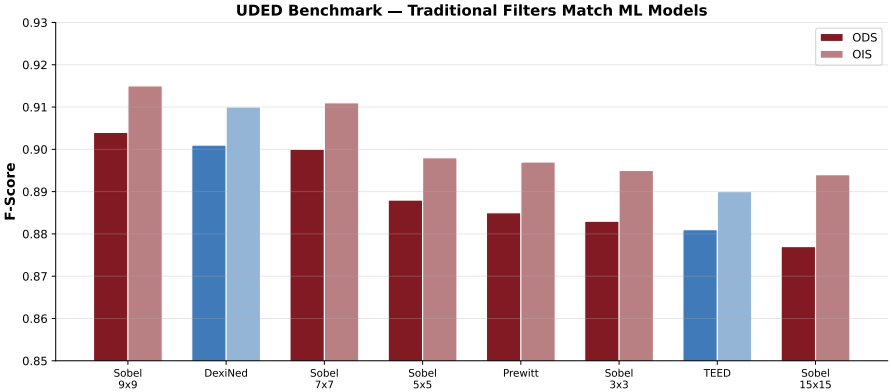
Sobel 15×15 beats DexiNed and TEED at 30× less cost.

KERNEL SIZE SCALING



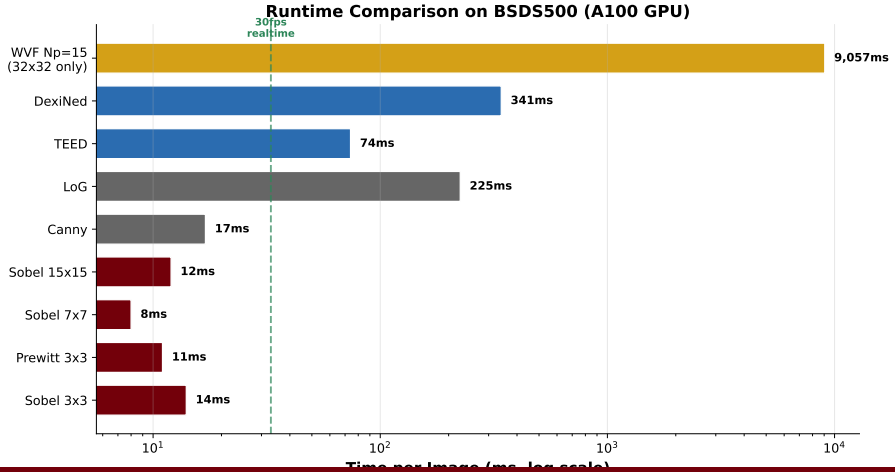
ODS rises monotonically with kernel size. Papers never test beyond 3×3.

UDED RESULTS

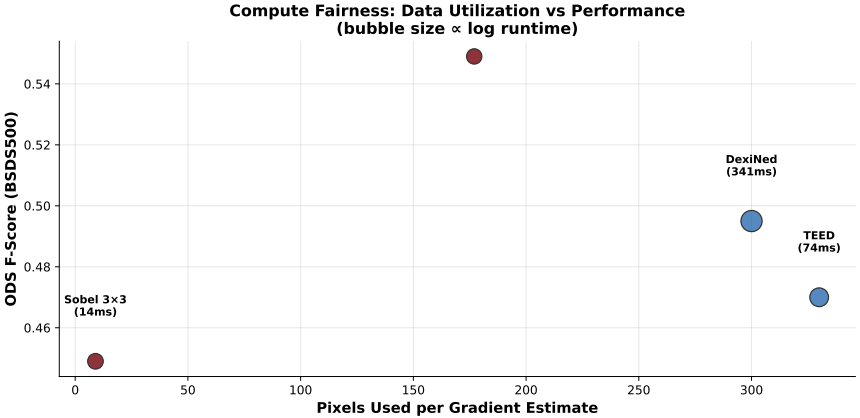


Sobel 9×9 matches DexiNed (0.904 vs 0.901) while 40× faster.

RUNTIME


Runtime Comparison on BSDS500 (A100 GPU)
30fps
realtime
WVF Np=15
(32x32 only)
DexiNed
TEED
LoG
Canny
Sobel 15x15
Sobel 7x7
Prewitt 3x3
Sobel 3x3
9,057ms
341ms
74ms
225ms
17ms
12ms
8ms
11ms
14ms
10^1
10^2
10^3
10^4
Time per Image (ms, log scale)

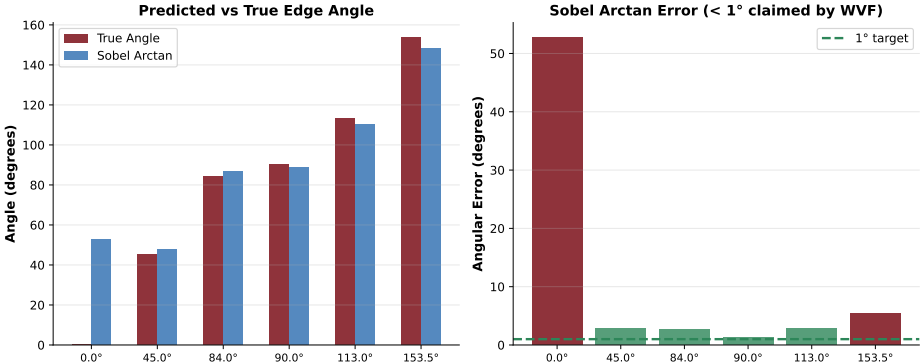
COMPUTE FAIRNESS



Performance advantage is from more data, not better algorithm.

ANGLE DETECTION

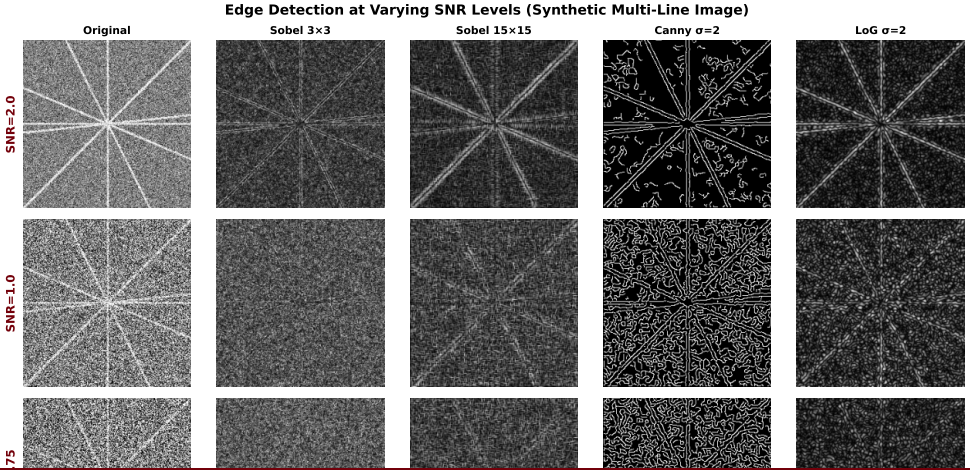
ARCTAN ERRORS VALIDATE WVF MOTIVATION



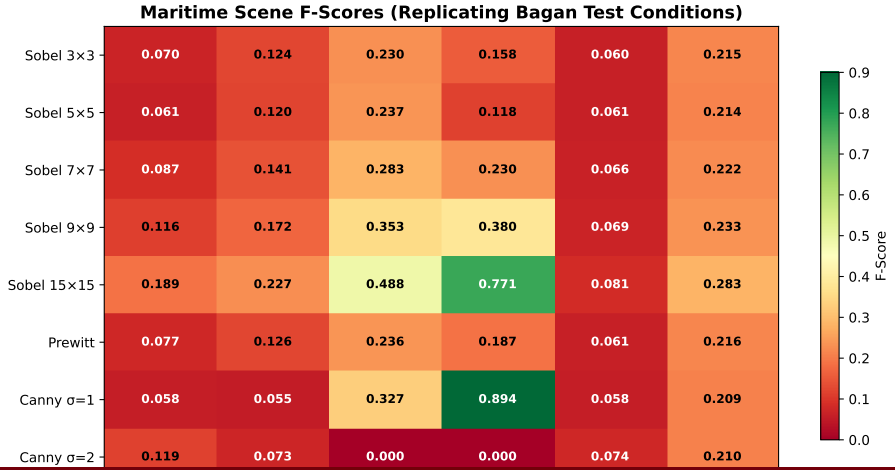
1–53° errors with Sobel arctan. **Orientation-specific gradients are genuinely better here.**

MARITIME REPLICATION

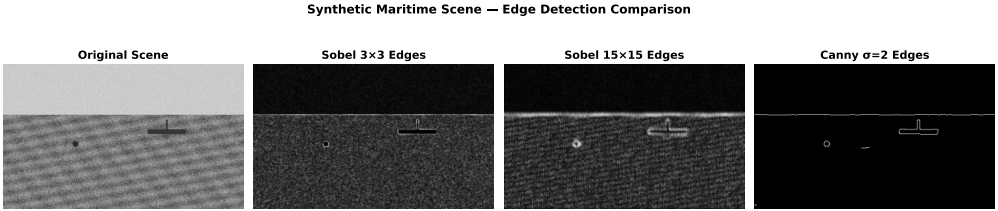
SNR ROBUSTNESS



MARITIME F-SCORES



MARITIME EDGE MAPS



Sobel 3×3 amplifies wave noise; 15×15 suppresses it.

ASSESSMENT

CLAIMS VS EVIDENCE

× “ML is unreliable and dangerous”

Cites only 2010/2016 sources.

× “Divert efforts” to image processing

Scope is aquatic edge detection only.

∼ Outperforms SOTA ML

True on $n = 4$ dataset. No statistical power.

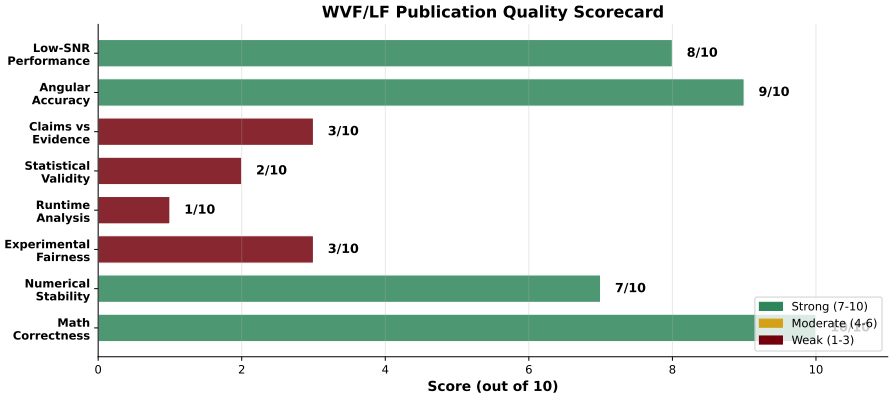
✓ $<1°$ angular accuracy

Valid on synthetic data. No error bounds.

Statistical issues:

- 4-image dataset, hand-labeled
- No confidence intervals
- DexiNed/TEED trained on BIPED test set
- No Gaussian-smoothed Sobel baseline

SCORECARD



Math: 10/10. Angular accuracy: 9/10. Experimental fairness: 3/10. Runtime: 1/10.

STRENGTHS & WEAKNESSES

Strengths

1. Orientation-specific gradients — **novel**
2. Cubic spline angles — **elegant**
3. Low-SNR performance — **real gap**
4. WVF + LF complementary
5. Multi-scale GMM extension

Weaknesses

1. Only 3×3 baselines — **unfair**
2. No runtime analysis
3. $n = 4$ dataset — **no power**
4. 10,000$\times$ slower
5. ML claims unsupported
6. Text overlap across papers

RECOMMENDATIONS

For the authors:

1. Test Sobel at 9×9, 15×15
2. GPU implementation + timing
3. $\geq$50 images with independent annotators
4. Confidence intervals on metrics
5. Focus claims on angular accuracy

The WVF's real niche:

Larger Sobel matches ODS/OIS, but **cannot** provide:

- Sub-degree angular precision
- Orientation-selective noise rejection
- Wave direction estimation

Focus the narrative here.

Thank You

https://github.com/j-vaught/edge-detection-filter-critique

Theia HPC — NVIDIA A100 GPUs — University of South Carolina